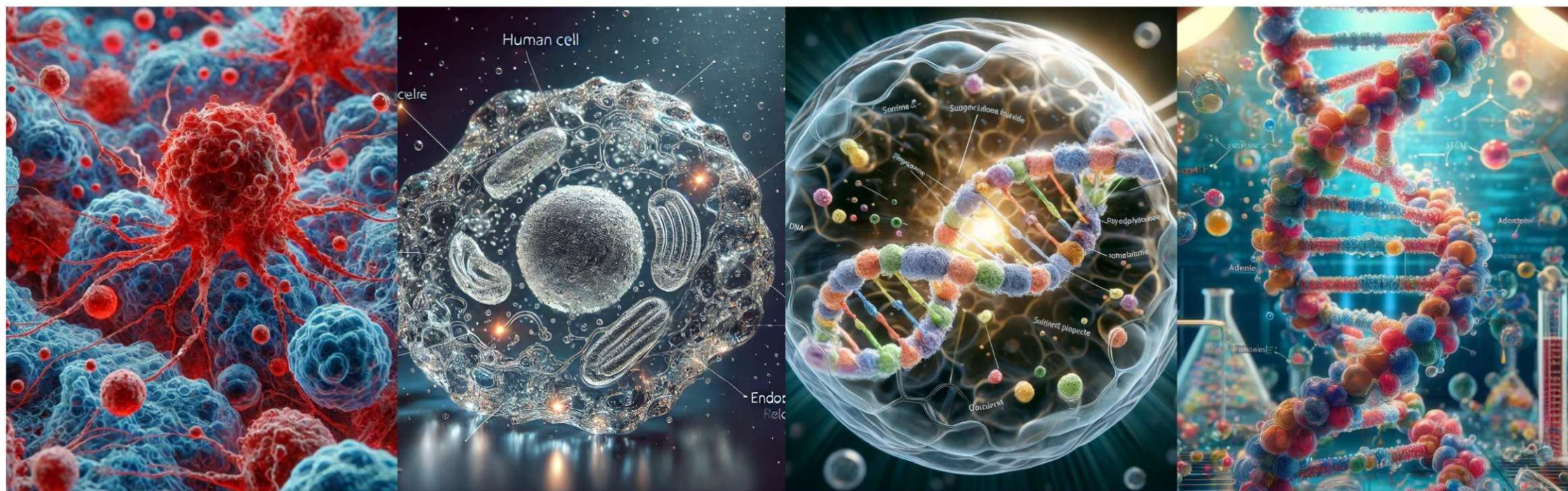


Omics-Based Clinical and Population Studies (session 1-2)

AI-Driven Patient Care

Transforming the Experience



Cancer Epi-omics cohort

Epi-omics And Drug Use: Identification Of Late Effects Trajectories

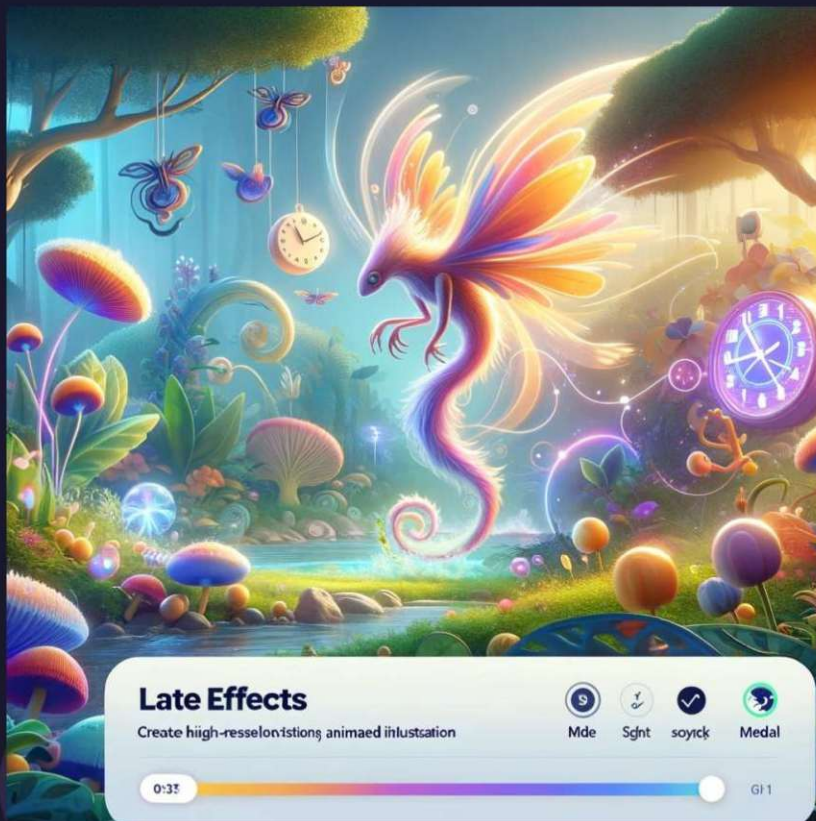
Variants of Uncertain Significance in Cancer and Patient Risk Profiles

SALICCS PROGRAMME EXPERIENCE: SCIENTIFIC COORDINATOR FOR THE PROGRAM

- My experience - the Nordic cross-border research programme SALiCCS, a collaboration with the Danish Cancer Society, the Finnish Cancer Registry, and Karolinska Institutet. This programme investigates late treatment effects and psychosocial conditions in childhood cancer survivors across Denmark, Finland, and Sweden.
- **Cohort Design:** SALiCCS is a registry-based matched cohort design including five-year survivors diagnosed between 1971-2011, along with age-, sex-, and country-matched comparisons and siblings.
- **Outcomes of Interest:** The programme examines morbidity and disease progression, mental disorders, education, employment, family life, and other related outcomes.
- **Data Linkage:** Data from national patient, cancer, and medication registries were used for individual-level data linkage.
- **Core Population:** The core population includes 21,292 five-year survivors, 103,303 population comparisons, and 29,644 siblings. Common diagnoses include central nervous system tumors, leukemias, and lymphomas.



Research domains: SALICCS: Nordic research projects



Treatments: Use of medications (whole profile)

- Compositions of treatments (chemo, radio, immunotherapies, etc.)
- Characterization of compositions (e.g., morbidity, stage of disease, demographics, etc.)

Quality of life: national survey

- Changes in life course: compositions of life events, activities, etc
- Social and health outcomes in survivors: compositions of patient socioeconomic and family context, trajectories of health late effects

COVID-19

- in children, young adults, and survivors of childhood cancer

Linking epi data to omics

- E.g., Children with leukemia (samples from healthy siblings' vs sick children)

Danish Data for my projects: DST, RKKP and SDS

Date Created: January 20, 2025
Last Modified: January 20, 2025

Purpose:

Link multiple Danish health registries including LPR2/LPR3 (patient registry), medication registry, vaccination registry, and population registry.

Input Datasets:

1. LPR.KONTAKTER - Hospital contacts from LPR3
2. LPR.DIAGNOSER - Diagnoses from LPR3
3. LPR.DIAG_INDL - Historical diagnoses from LPR2
4. MED.LMDB202312 - Medication data (December 2023 extract)
5. MED.LMDB202306 - Medication data (June 2023 extract)
6. VAC.EPIDOV_VACCINEDATA - Vaccination data
7. BEF.BEF202312 - Population registry (December 2023 extract)

Output Datasets:

1. WORK.FINAL_COHORT - Final linked dataset
2. WORK.PATIENT_TIMELINE - Detailed patient event timeline

Notes:

- LPR2 data cutoff: 30NOV2013
- LPR3 contains data from 2018/2019 onwards
- Population database contains only living citizens
- Multiple records per patient exist in all datasets except population

Indholdsfortegnelse

- 1 [Intro](#)
- 2 [Data flow](#)
 - 2.1 [Data kilder](#)
 - 2.2 [Data ansøgning](#)
- 3 [Hvordan læser jeg dokumentet?](#)
- 4 [Befolkningen](#)
 - 4.1 [Det centrale personregister \(CPR-registeret\)](#)
- 5 [Ind-/udvandring](#)
- 6 [Død og dødsårsager](#)
- 7 [Cancerregisteret](#)
- 8 [Uddannelse](#)
- 9 [Indkomst](#)
- 10 [Arbejdsmarkedstilknøytning](#)
- 11 [Landspatientregisteret](#)
 - 11.1 [LPR2](#)
 - 11.1.1 [Indlæggelser](#)
 - 11.1.2 [Privathospitaler](#)
 - 11.1.3 [Psykriske indlæggelser](#)
 - 11.1.4 [Fødsler fra LPR](#)
 - 11.1.5 [Medicinsk fødselsregister](#)
 - 11.1.6 [Fertilitetsdatabasen](#)
 - 11.2 [LPR3_F](#)
- 12 [Register over In Vitro Fertilitetsbehandling \(IVF\)](#)
- 13 [Landsdækkende Register for Patologi](#)
- 14 [Det Psykiatriske Centrale Forskningsregister](#)
- 15 [Sygesikringsregisteret](#)
- 16 [Register for Udvalgte Kroniske Sygdomme og svære psykiske lidelser](#)
- 17 [Lægemedladata](#)
 - 17.1 [Lægemedladedatabasen](#)
 - 17.2 [Sygehusmedicinregisteret](#)
- 18 [Laboratoriedata](#)

@NerminGhith

Variabelliste for DST

KB projektdatabase 704075

5/26/24, 11:56 PM

706024_project proposal_20240526_v2

APPENDIX 1 – Appendix documents available in application

Project proposal 706024 (2):
Adult Life after Childhood
Cancer in Scandinavia(2018-
DCRC-0044)

FILE NAME	ALIAS	ATTACHED TO	DATE
Additional data from Denmark_ng.pdf		Data fra projektdatabasen (704075) (POP B)	06.12.2023
Additional data from Sweden.pdf		Nordic data (POP B)	03.11.2023
Additional variables from Finland.pdf		Nordic data (POP B)	03.11.2023
EpiDDV_Vaccinedata.xlsx		Data from SDS (POP B)	08.12.2023
New 2023-05-01_Variabeloversigt_DBCR.XLSX		Data from RKKP (POP B)	05.12.2023
New FOMY Parental custody.docx		Data fra projektdatabasen (704075) (POP B)	23.09.2023
New RKKP Variabeloversigt_DACOVID.xlsx		Data from RKKP (POP B)	27.10.2023
New RKKP Variabeloversigt_DID.xlsx		Data from RKKP (POP B)	27.10.2023
New SDS covid-19 testdata pr. case.xlsx		Data from SDS (POP B)	07.12.2023
New__LPR.xlsx		Data fra projektdatabasen (704075) (POP B)	06.12.2023
New_LMDB.xlsx		Data from SDS (POP B)	06.12.2023
Parental information.xlsx		Data fra projektdatabasen (704075) (POP B)	05.12.2023
Variables on cohabitation, leaving home, marital status.xlsx		Data fra projektdatabasen (704075) (POP B)	05.12.2023
Variables on education.xlsx		Data fra projektdatabasen (704075) (POP B)	05.12.2023
Variables on occupation, work, income.xlsx		Data fra projektdatabasen (704075) (POP B)	05.12.2023
Variables on parenthood.xlsx		Data fra projektdatabasen (704075) (POP B)	05.12.2023
Variables on social security benefits.xlsx		Data fra projektdatabasen (704075) (POP B)	05.12.2023

28-01-2025



Drug use

Use of prescription drugs among childhood cancer survivors – a population-based study of late effects in Nordic countries

MAPPING GENERAL USE

The overall use of prescription drugs and prevalence of polypharmacy in childhood cancer survivors including all fourteen main groups of prescription drugs and subgroups (Study 1)



ANTIBIOTICS

The use of antibiotics in childhood cancer survivors compared to that in the background population (Study 2)



NERVOUS SYS. DRUGS

The use of analgesics, antidepressants, and antipsychotics prescribed and especially potent analgesics or opioids in childhood cancer survivors compared to that in the background population (Study 3)



Own data

Figure 1. Rate ratios and 95% CIs for prescriptions within the 65 ATC 2nd level groups of prescription drugs (dispensed more than 100 times among survivors), comparing survivors of childhood cancer with population comparisons

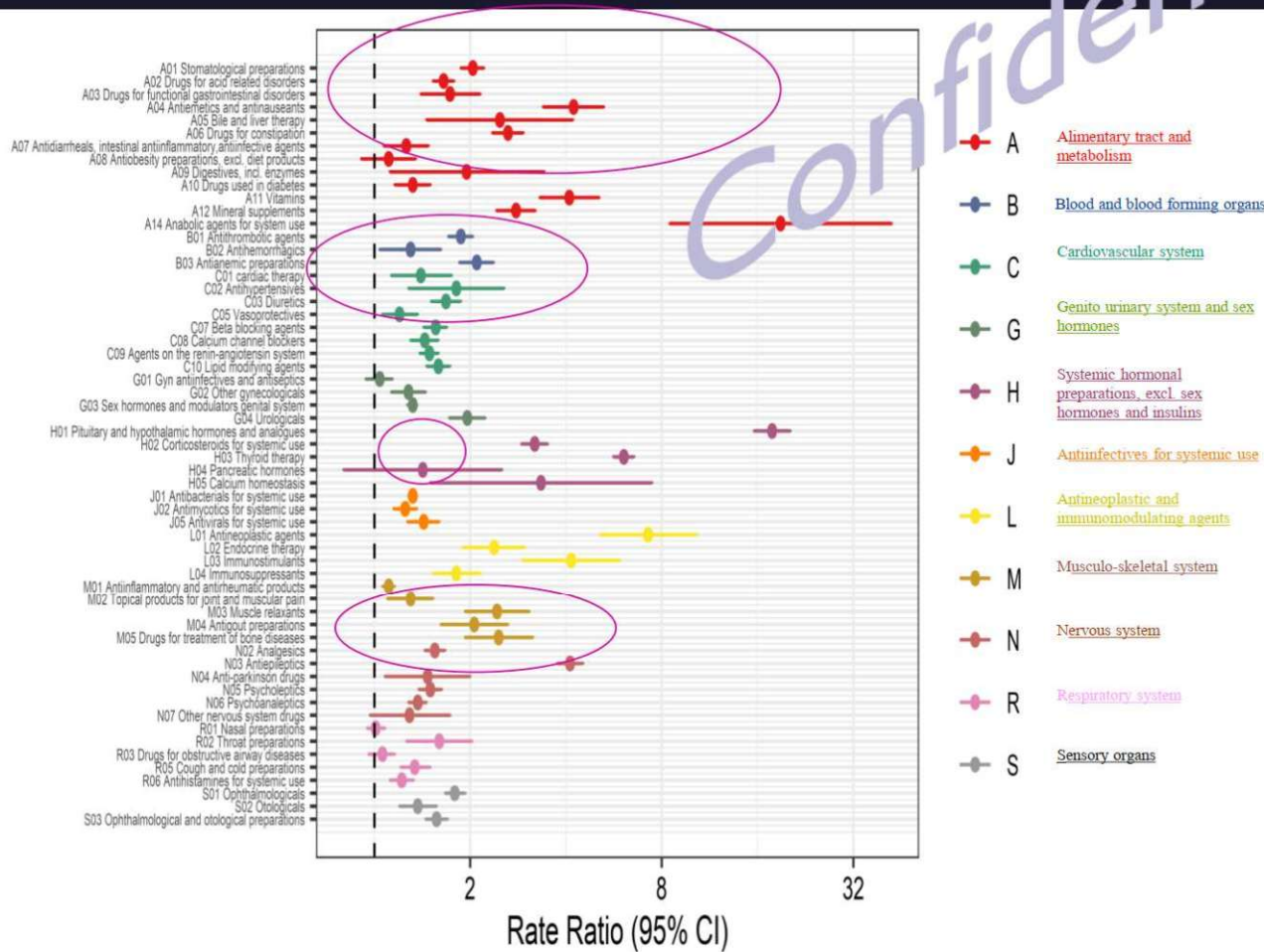
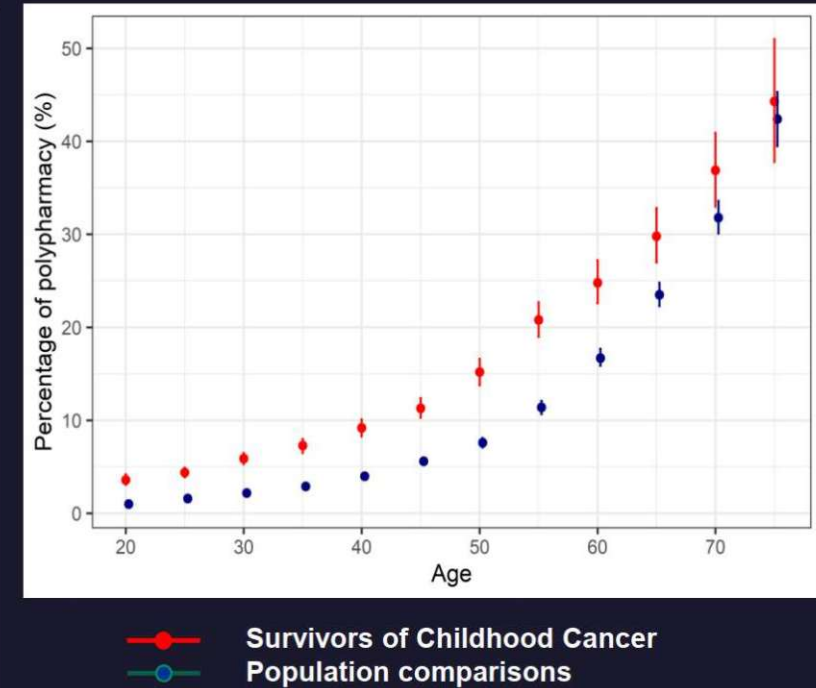


Figure 2. Prevalence and 95% confidence intervals in childhood cancer survivors and population comparisons with polypharmacy



Proposed new cohort: Linking Epi-omics and Drug Data For The Identification Of Late Effects Trajectories

The Recent Research Proposals: Gene Variants In Childhood Cancer

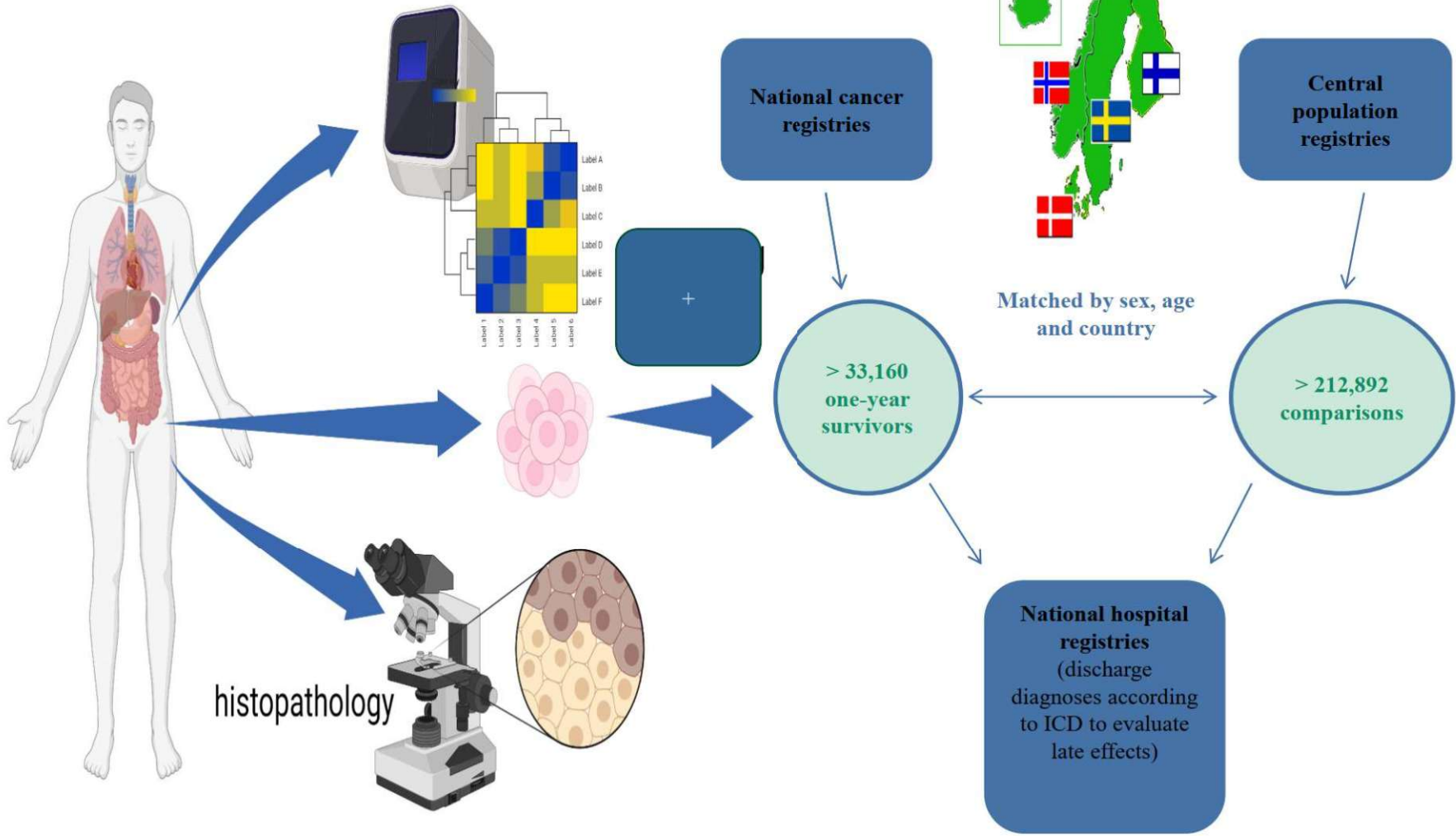
Key Techniques and Approaches

- **Large-Scale Record Linkage:** Utilizing national health outcome registers in nationwide population-based settings to link various large-scale records.
- **Characterized Variants and Clinical Data:** Leveraging data from registries to enhance understanding of tumor evolution, triggers, and risk factors for second cancers and late effects and improve accuracy in treatment response predictions and survival estimates.
- **Comprehensive Dataset Development:** Using national and regional cancer registries to develop a comprehensive dataset on children, teenagers, and young adults with cancer.
- **Population-Level Analysis:** Conducting large-scale analyses to explore the relationship between genetic variants and the onset of cancer, survival, treatment resistance, and late effects.
- **Follow-Up Health Data:** Utilizing medical records, prescription and medication details, and follow-up health data from national registries for survivor comparisons.



Own data

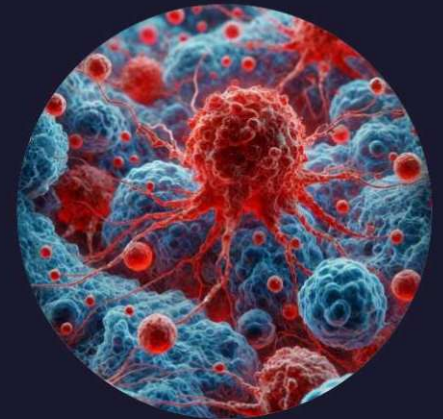
NGS analysis



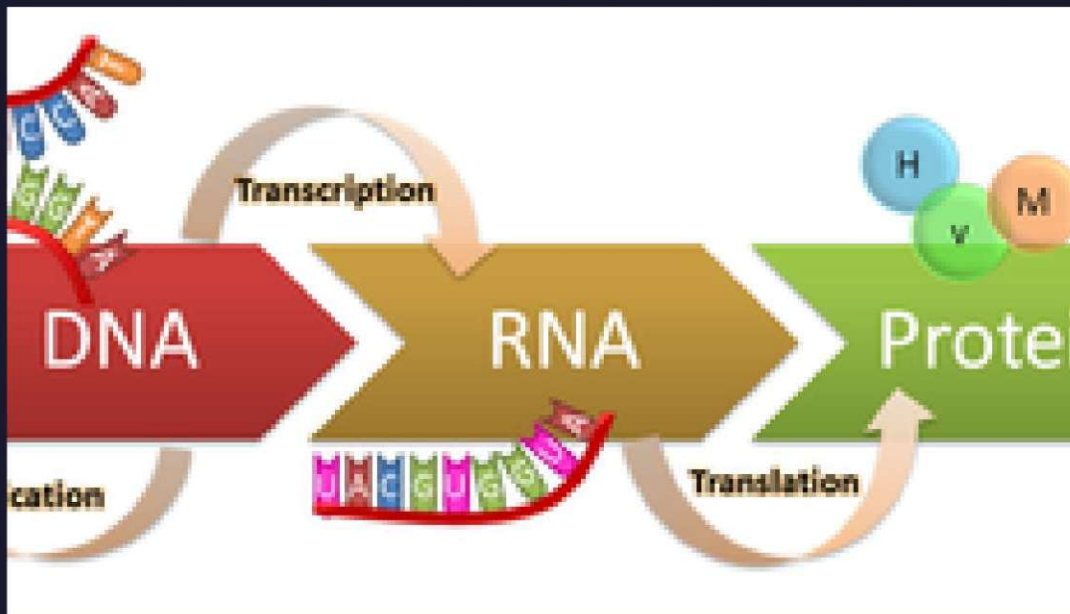
New Epi. Cohort to Investigate: Variants Of Uncertain Significance In Rare Types Of Cancer

Characterization and **stratification** of patient morbidity clusters, risk of secondary cancer, prognosis, and survival

- Link **transcriptomic** data with clinical and population data in the national registries.
- Compare patients with their **siblings** and the **general** population



Transcriptomics



- “Transcriptomics is the study of the structure, function, and evolution of the transcriptome (i.e., the entirety of RNA transcripts produced by the genome) of a given organism or community of organisms under a variety of conditions”.
- “An RNA transcript is the RNA strand that is produced when a gene is transcribed.”

Sources: <https://toolkit.ncats.nih.gov/glossary/rna-transcript/#:~:text=An%20RNA%20transcript%20is%20the,is%20translated%20into%20a%20protein.>

<https://www.mrgscience.com/topic-27-dna-replication-transcription-and-translation.html>

<https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/transcriptomics>

VUS definition:
“A change in a **gene's DNA sequence** that has an unknown effect on a person's health. There is usually not enough information about a variant of uncertain significance to know whether it increases a person's risk of developing a disease, such as cancer. Also called unclassified variant, variant of unknown significance, and VUS.”

Source:
<https://www.cancer.gov/publications/dictionaries/cancer-terms/def/variant-of-uncertain-significance>



Created in BioRender.com 

Figure source: <https://curesyngap1.org/blog/does-your-genetic-report-contain-a-variant-of-unknown-significance-vus-in-syngap1/>

Source:

Precision Nutrition for Cardiovascular Disease Prevention

- January 2023
- [Lifestyle Genomics](#) 16(1)
- DOI:
- [10.1159/000529054](#)
- License
- [CC BY-NC 4.0](#)

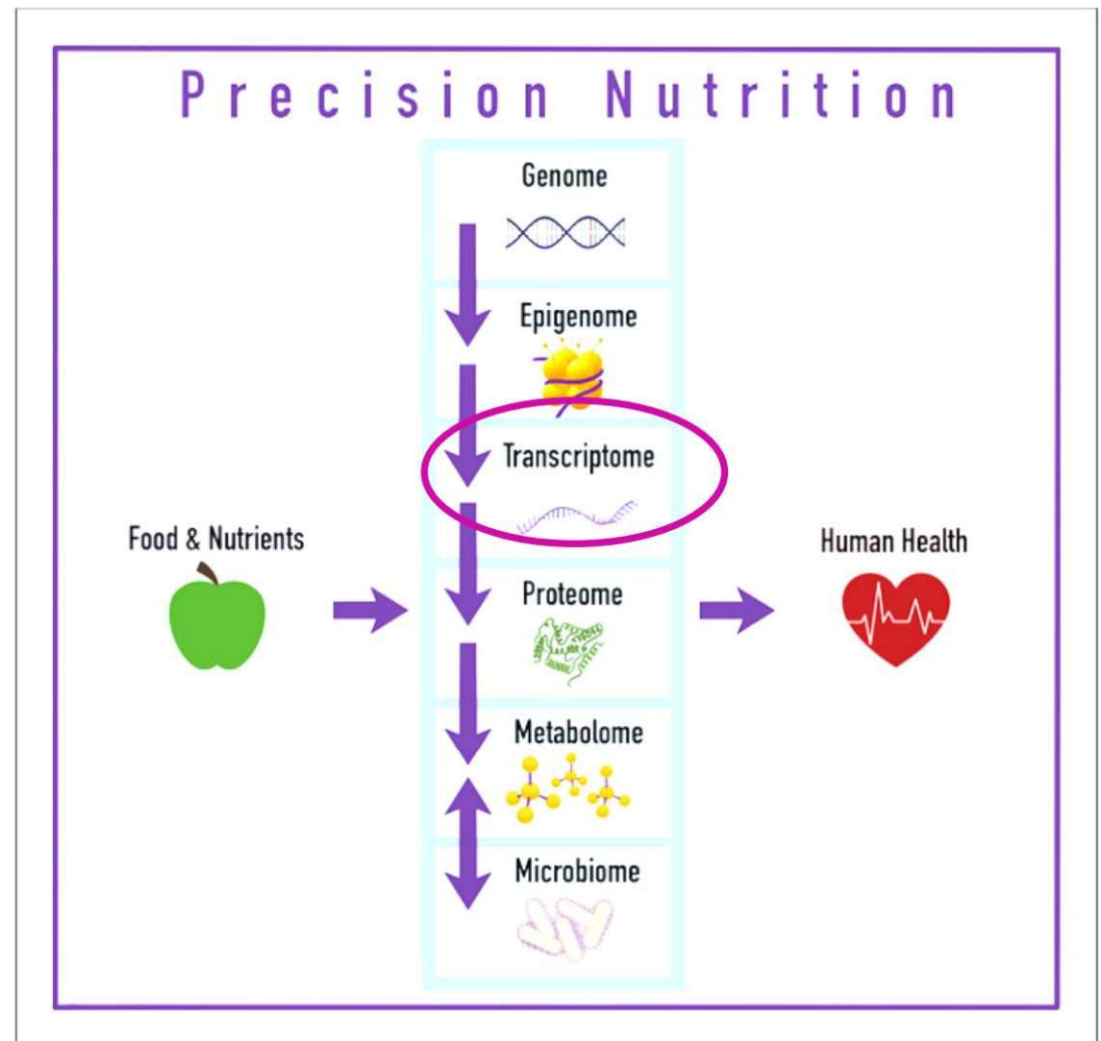


Fig. 1. Schematic representation of the main components of precision nutrition influencing human health.

Research domains: SALICCS Nordic research projects



Childhood Cancer Research Group (CCA)
Transcending Boundaries: From Clinical and Medical Labs to
Population-Based Research in Childhood
Cancer Survivors



Use of prescription drugs in Danish and Swedish childhood cancer survivors: A population-based cohort study

Nermin Ghith and Jesper E. Winkler, MD, DMSc*
* On behalf of SALICCS research program collaborators



Research Group for Childhood Cancer
Danish Cancer Society Research Center
Copenhagen, 2300 S, DK-2300
Research Center for Childhood Cancer
Danish Cancer Society Research Center

Prescription date: 18 June 2023
Presentation date: 18 June 2023

Aim

The aim of this cohort study was to provide a detailed yet comprehensive overview of the use of prescription drugs in childhood cancer survivors diagnosed in Denmark and Sweden as well as to investigate the prevalence of polypharmacy.

Methods

Based on unique data from the Nordic population-based research program SALICCS, all children with cancer before the age of 20 years in Denmark (1945-2008) and Sweden (1965-2011) were identified. The study population consisting of 16,742 five-year childhood cancer survivors identified in the nationwide cancer registries and classified according to the International Classification of Childhood Cancer, with 77,701 population comparisons (without cancer) randomly selected from the population registers and 23,105 sibling comparisons. Information on drug prescriptions was obtained by linking the three cohorts to the nationwide prescription registries in Denmark and Sweden with information on all drugs dispensed at pharmacies using the hierarchical Anatomical Therapeutic Chemical (ATC) classification system. For each dispensing, information was obtained on the ATC-code at 5th level (chemical substance) and date of dispensing. All prescriptions were grouped into 11 ATC 1st main level and 65 ATC 2nd level therapeutic groups of prescription drugs. Rates ratios (RRs) and 95% confidence intervals (CIs) for filled drug prescriptions were calculated in marginal rates models allowing each individual to have several prescriptions during follow-up. Relative rates of prescriptions were calculated for the 11 ATC 1st main level and each of the 65 ATC 2nd level therapeutic groups of prescription drugs.

Table 1. Rate ratios (RR) and 95% CIs for all childhood cancer survivors and for the subgroups of histological malignancies, CNS tumors or non-CNS tumors compared to population and sibling comparisons.

	All childhood cancer survivors	Histological malignancies	CNS tumors	Non-CNS tumors
RR (95% CI)	RR (95% CI)	RR (95% CI)	RR (95% CI)	RR (95% CI)
Survivors of childhood cancer (population comparisons)	1.35 (1.32-1.38)	1.48 (1.42-1.54)	1.38 (1.30-1.47)	1.35 (1.32-1.41)
1 (ref)	1 (ref)	1 (ref)	1 (ref)	1 (ref)
Siblings	1.76 (1.69-1.82)	1.65 (1.57-1.73)	2.38 (2.23-2.53)	1.48 (1.39-1.58)
Age at diagnosis/index date				
0-4	1.64 (1.49-1.81)	1.85 (1.40-2.34)	2.37 (1.90-2.92)	1.45 (1.27-1.63)
5-9	1.64 (1.54-1.72)	1.49 (1.38-1.61)	2.18 (1.99-2.39)	1.51 (1.40-1.64)
10-14	1.62 (1.54-1.71)	1.33 (1.23-1.43)	2.15 (1.97-2.34)	1.35 (1.23-1.51)
15-19	1.58 (1.52-1.65)	1.44 (1.31-1.58)	1.97 (1.81-2.14)	1.37 (1.27-1.48)
20-24	1.45 (1.39-1.52)	1.58 (1.46-1.71)	1.75 (1.63-1.88)	1.30 (1.20-1.38)
Time since diagnosis/index date				
0-4 years	1.38 (1.37-1.40)	1.79 (1.68-1.90)	2.54 (2.38-2.72)	1.68 (1.58-1.77)
5-9 years	1.61 (1.56-1.66)	1.49 (1.42-1.57)	2.10 (1.98-2.23)	1.43 (1.36-1.50)
10-14 years	1.48 (1.44-1.53)	1.42 (1.34-1.50)	1.88 (1.78-1.98)	1.31 (1.26-1.37)
15-19 years	<.0001	<.0001	<.0001	<.0001
Survivors of childhood cancer (sibling comparisons)	1.60 (1.58-1.63)	1.59 (1.52-1.67)	2.09 (1.97-2.22)	1.39 (1.33-1.45)
1 (ref)	1 (ref)	1 (ref)	1 (ref)	1 (ref)
Age at diagnosis/index date				
0-4	1.82 (1.79-1.85)	1.77 (1.65-1.90)	2.36 (2.14-2.60)	1.55 (1.42-1.68)
5-9	1.48 (1.42-1.53)	1.44 (1.33-1.55)	1.88 (1.74-2.04)	1.32 (1.22-1.38)
Time since diagnosis/index date				
0-4 years	1.64 (1.70-1.57)	1.79 (1.66-1.93)	2.34 (2.18-2.51)	1.58 (1.49-1.67)
5-9 years	1.58 (1.56-1.60)	1.48 (1.42-1.55)	2.08 (1.98-2.18)	1.38 (1.32-1.43)
10-14 years	1.41 (1.38-1.44)	1.41 (1.30-1.52)	2.08 (1.95-2.24)	1.37 (1.30-1.45)
15-19 years	1.61 (1.55-1.66)	1.61 (1.50-1.72)	1.85 (1.75-1.94)	1.47 (1.38-1.56)

* Adjusted for country, sex and year group at diagnosis/index date with age as the underlying time scale.
* Test for effect: Test of whether the exposure effect is the same over the three groups of years since diagnosis/index date.

Photo credit: graphish website

Results

Survivors had a 50-60% higher rate of drug prescriptions (table 1) compared with comparisons (RR 1.55; 95% CI 1.52-1.58) and siblings (RR 1.32; 95% CI 1.50-1.67). Strikingly, three overall cancer groups (table 1), the highest rate of prescriptions was seen for survivors of CNS tumors (RR 1.58; 95% CI 1.52-1.67) followed by survivors of hematological malignancies (RR 1.48; 95% CI 1.42-1.54) and non-CNS tumors (RR 1.35; 95% CI 1.32-1.41) compared with the population comparisons.

The relative rates of polypharmacy for survivors compared with population comparisons were higher in all 11 ATC 1st level and most of the 65 ATC 2nd level groups of prescription drugs (figure 1) with the highest rates seen for systematic hormonal preparations and antineoplastic and immunomodulatory agents (at the ATC 1st level).

The prevalence of polypharmacy increased with age in both survivors and population comparisons (figure 2). At all ages until 70 years, a significantly higher proportion of survivors filled five or more prescriptions during a 120-day period compared with population comparisons, particularly for CNS tumor survivors (see figure 2 and table 2).

Conclusions

Our findings emphasize the need for increased awareness for this vulnerable group of childhood cancer survivors at high risk for adverse health outcomes resulting from multiple chronic medications.

Figure 1. Rate ratios (RR) and 95% CIs for prescriptions within the 65 ATC 2nd level groups of prescription drugs (dispensed more than 100 times among survivors), comparing survivors of childhood cancer with population comparisons.

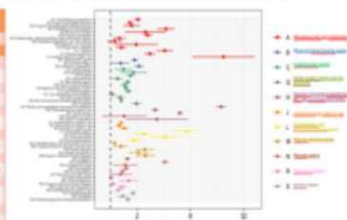


Figure 2. Prevalence and 95% confidence intervals in childhood cancer survivors and population comparisons with polypharmacy.

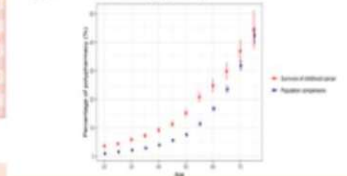


Table 2. The number of persons and the proportion with polypharmacy within the 120 days before age 30, 40, 50 and 60, stratified by cancer type.

	Numbers of persons	Proportion with polypharmacy	Numbers of persons	Proportion with polypharmacy
	N	% (95% CI)	N	% (95% CI)
Age 30			Age 40	
Polymorphous connective tissue diseases	25,179	2.2 (2.0-2.4)	35,345	4.0 (3.8-4.2)
Endocrine disorders	1,111	4.0 (3.7-4.3)	567	8.2 (6.2-10.0)
Immunodeficiency	813	4.5 (3.2-6.0)	460	8.9 (6.8-11.4)
CNS tumors	1,387	8.9 (7.4-10.5)	1,025	15.1 (13.4-17.1)
Hematological malignancies	304	5.8 (3.3-12.2)	62	11.3 (4.7-21.9)
Non-hematological malignancies	147	2.7 (0.7-4.6)	156	6.4 (3.1-11.5)
Neural tumors	239	5.9 (3.2-9.6)	185	9.2 (5.4-14.3)
Other tumors	17	5.9 (0.1-28.7)	-	-
Other tumors	179	8.1 (4.5-12.2)	148	8.0 (5.3-10.4)
Soft tissue sarcoma	268	3.7 (1.4-6.0)	192	8.1 (4.8-13.2)
Soft tissue sarcoma	475	4.2 (2.4-6.4)	386	6.9 (4.2-9.4)
Soft tissue sarcoma	174	3.5 (1.4-5.3)	387	5.9 (3.6-7.7)
Soft tissue sarcoma	82	3.7 (0.8-10.0)	151	8.4 (4.9-14.5)
Soft tissue sarcoma	<0.0001	<0.0001	<0.0001	<0.0001
Age 50			Age 60	
Polymorphous connective tissue diseases	11,087	7.4 (7.1-7.7)	6,109	18.2 (15.9-21.7)
Endocrine disorders	147	11.6 (6.9-17.0)	31	25.6 (13.9-44.0)
Immunodeficiency	174	15.0 (11.5-19.0)	185	28.2 (21.8-35.3)
CNS tumors	681	28.9 (26.3-32.1)	387	28.2 (23.7-32.9)
Hematological malignancies	35	6.6 (3.9-11.2)	11	36.4 (10.9-69.2)
Non-hematological malignancies	118	4.6 (2.0-12.0)	46	21.7 (10.9-36.4)
Neural tumors	89	10.1 (5.7-17.4)	21	23.8 (14.7-37.0)
Other tumors	99	15.2 (8.7-23.8)	64	23.4 (13.8-35.7)
Soft tissue sarcoma	336	8.8 (5.3-13.9)	116	19.1 (12.4-27.6)
Soft tissue sarcoma	217	16.7 (12.4-21.9)	107	23.4 (15.7-32.5)
Soft tissue sarcoma	465	14.0 (10.6-17.9)	333	21.4 (13.8-30.4)
Soft tissue sarcoma	113	15.0 (9.0-23.0)	85	25.9 (17.0-36.1)
Soft tissue sarcoma	<0.0001	<0.0001	<0.0001	<0.0001

* Polypharmacy was defined as receiving more than five different prescription medications during 120 days before an age.
* The P-value is from a chi-squared test comparing the proportion with polypharmacy in the 12 cancer groups and the population comparisons.

Childhood Cancer Research Group (CCA)
Transcending Boundaries: From Clinical and Medical Labs to
Population-Based Research in Childhood
Cancer Survivors

Survivorship and drug AI data lab: Future is here*

Our research has undergone various phases of transformation, progressing from the utilization of information found in clinical records and databases to establishing connections and integrating evidence through extensive natural and regional registries.

Today, we stand at the forefront of innovation, harnessing the power of data science tools to shape an AI lab. Our mission is to comprehensively explore the diverse, interdisciplinary dimensions of Childhood Cancer by seamlessly integrating data on late effects, treatments, as well as biological and psychosocial factors.

Contact Nermin Ghith for more information

1. Life after childhood cancer: a population-based cohort study

2. Life after childhood cancer: a population-based cohort study

3. Life after childhood cancer: a population-based cohort study

4. Life after childhood cancer: a population-based cohort study

5. Life after childhood cancer: a population-based cohort study

6. Life after childhood cancer: a population-based cohort study

7. Life after childhood cancer: a population-based cohort study

8. Life after childhood cancer: a population-based cohort study

9. Life after childhood cancer: a population-based cohort study

10. Life after childhood cancer: a population-based cohort study

11. Life after childhood cancer: a population-based cohort study

12. Life after childhood cancer: a population-based cohort study

13. Life after childhood cancer: a population-based cohort study

14. Life after childhood cancer: a population-based cohort study

15. Life after childhood cancer: a population-based cohort study

16. Life after childhood cancer: a population-based cohort study

17. Life after childhood cancer: a population-based cohort study

18. Life after childhood cancer: a population-based cohort study

19. Life after childhood cancer: a population-based cohort study

20. Life after childhood cancer: a population-based cohort study

21. Life after childhood cancer: a population-based cohort study

22. Life after childhood cancer: a population-based cohort study

23. Life after childhood cancer: a population-based cohort study

24. Life after childhood cancer: a population-based cohort study

25. Life after childhood cancer: a population-based cohort study

26. Life after childhood cancer: a population-based cohort study

27. Life after childhood cancer: a population-based cohort study

28. Life after childhood cancer: a population-based cohort study

29. Life after childhood cancer: a population-based cohort study

30. Life after childhood cancer: a population-based cohort study

31. Life after childhood cancer: a population-based cohort study

32. Life after childhood cancer: a population-based cohort study

33. Life after childhood cancer: a population-based cohort study

34. Life after childhood cancer: a population-based cohort study

35. Life after childhood cancer: a population-based cohort study

36. Life after childhood cancer: a population-based cohort study

37. Life after childhood cancer: a population-based cohort study

38. Life after childhood cancer: a population-based cohort study

39. Life after childhood cancer: a population-based cohort study

40. Life after childhood cancer: a population-based cohort study

41. Life after childhood cancer: a population-based cohort study

42. Life after childhood cancer: a population-based cohort study

43. Life after childhood cancer: a population-based cohort study

44. Life after childhood cancer: a population-based cohort study

45. Life after childhood cancer: a population-based cohort study

46. Life after childhood cancer: a population-based cohort study

47. Life after childhood cancer: a population-based cohort study

48. Life after childhood cancer: a population-based cohort study

49. Life after childhood cancer: a population-based cohort study

50. Life after childhood cancer: a population-based cohort study

51. Life after childhood cancer: a population-based cohort study

52. Life after childhood cancer: a population-based cohort study

53. Life after childhood cancer: a population-based cohort study

54. Life after childhood cancer: a population-based cohort study

55. Life after childhood cancer: a population-based cohort study

56. Life after childhood cancer: a population-based cohort study

57. Life after childhood cancer: a population-based cohort study

58. Life after childhood cancer: a population-based cohort study

59. Life after childhood cancer: a population-based cohort study

60. Life after childhood cancer: a population-based cohort study

61. Life after childhood cancer: a population-based cohort study

62. Life after childhood cancer: a population-based cohort study

63. Life after childhood cancer: a population-based cohort study

64. Life after childhood cancer: a population-based cohort study

65. Life after childhood cancer: a population-based cohort study

66. Life after childhood cancer: a population-based cohort study

67. Life after childhood cancer: a population-based cohort study

68. Life after childhood cancer: a population-based cohort study

69. Life after childhood cancer: a population-based cohort study

70. Life after childhood cancer: a population-based cohort study

71. Life after childhood cancer: a population-based cohort study

72. Life after childhood cancer: a population-based cohort study

73. Life after childhood cancer: a population-based cohort study

74. Life after childhood cancer: a population-based cohort study

75. Life after childhood cancer: a population-based cohort study

76. Life after childhood cancer: a population-based cohort study

77. Life after childhood cancer: a population-based cohort study

78. Life after childhood cancer: a population-based cohort study

79. Life after childhood cancer: a population-based cohort study

80. Life after childhood cancer: a population-based cohort study

81. Life after childhood cancer: a population-based cohort study

82. Life after childhood cancer: a population-based cohort study

83. Life after childhood cancer: a population-based cohort study

84. Life after childhood cancer: a population-based cohort study

85. Life after childhood cancer: a population-based cohort study

86. Life after childhood cancer: a population-based cohort study

87. Life after childhood cancer: a population-based cohort study

88. Life after childhood cancer: a population-based cohort study

89. Life after childhood cancer: a population-based cohort study

90. Life after childhood cancer: a population-based cohort study

91. Life after childhood cancer: a population-based cohort study

92. Life after childhood cancer: a population-based cohort study

93. Life after childhood cancer: a population-based cohort study

94. Life after childhood cancer: a population-based cohort study

95. Life after childhood cancer: a population-based cohort study

96. Life after childhood cancer: a population-based cohort study

97. Life after childhood cancer: a population-based cohort study

98. Life after childhood cancer: a population-based cohort study

99. Life after childhood cancer: a population-based cohort study

100. Life after childhood cancer: a population-based cohort study

101. Life after childhood cancer: a population-based cohort study

102. Life after childhood cancer: a population-based cohort study

103. Life after childhood cancer: a population-based cohort study

104. Life after childhood cancer: a population-based cohort study

105. Life after childhood cancer: a population-based cohort study

106. Life after childhood cancer: a population-based cohort study

107. Life after childhood cancer: a population-based cohort study

108. Life after childhood cancer: a population-based cohort study

109. Life after childhood cancer: a population-based cohort study

110. Life after childhood cancer: a population-based cohort study

111. Life after childhood cancer: a population-based cohort study

112. Life after childhood cancer: a population-based cohort study

113. Life after childhood cancer: a population-based cohort study

114. Life after childhood cancer: a population-based cohort study

115. Life after childhood cancer: a population-based cohort study

116. Life after childhood cancer: a population-based cohort study

117. Life after childhood cancer: a population-based cohort study

118. Life after childhood cancer: a population-based cohort study

119. Life after childhood cancer: a population-based cohort study

120. Life after childhood cancer: a population-based cohort study

121. Life after childhood cancer: a population-based cohort study

122. Life after childhood cancer: a population-based cohort study

123. Life after childhood cancer: a population-based cohort study

124. Life after childhood cancer: a population-based cohort study

125. Life after childhood cancer: a population-based cohort study

126. Life after childhood cancer: a population-based cohort study

127. Life after childhood cancer: a population-based cohort study

128. Life after childhood cancer: a population-based cohort study

129. Life after childhood cancer: a population-based cohort study

130. Life after childhood cancer: a population-based cohort study

131. Life after childhood cancer: a population-based cohort study

132. Life after childhood cancer: a population-based cohort study

133. Life after childhood cancer: a population-based cohort study

134. Life after childhood cancer: a population-based cohort study

135. Life after childhood cancer: a population-based cohort study

136. Life after childhood cancer: a population-based cohort study

137. Life after childhood cancer: a population-based cohort study

138. Life after childhood cancer: a population-based cohort study

139. Life after childhood cancer: a population-based cohort study

140. Life after childhood cancer: a population-based cohort study

141. Life after childhood cancer: a population-based cohort study

142. Life after childhood cancer: a population-based cohort study

143. Life after childhood cancer: a population-based cohort study

144. Life after childhood cancer: a population-based cohort study

145. Life after childhood cancer: a population-based cohort study

146. Life after childhood cancer: a population-based cohort study

147. Life after childhood cancer: a population-based cohort study

148. Life after childhood cancer: a population-based cohort study

149. Life after childhood cancer: a population-based cohort study

150. Life after childhood cancer: a population-based cohort study

151. Life after childhood cancer: a population-based cohort study

152. Life after childhood cancer: a population-based cohort study

153. Life after childhood cancer: a population-based cohort study

154. Life after childhood cancer: a population-based cohort study

155. Life after childhood cancer: a population-based cohort study

156. Life after childhood cancer: a population-based cohort study

157. Life after childhood cancer: a population-based cohort study

158. Life after childhood cancer: a population-based cohort study

159. Life after childhood cancer: a population-based cohort study

160. Life after childhood cancer: a population-based cohort study

161. Life after childhood cancer: a population-based cohort study

162. Life after childhood cancer: a population-based cohort study

163. Life after childhood cancer: a population-based cohort study

164. Life after childhood cancer: a population-based cohort study

165. Life after childhood cancer: a population-based cohort study

166. Life after childhood cancer: a population-based cohort study

167. Life after childhood cancer: a population-based cohort study

168. Life after childhood cancer: a population-based cohort study

169. Life after childhood cancer: a population-based cohort study

170. Life after childhood cancer: a population-based cohort study

171. Life after childhood cancer: a population-based cohort study

172. Life after childhood cancer: a population-based cohort study

173. Life after childhood cancer: a population-based cohort study

174. Life after childhood cancer: a population-based cohort study

175. Life after childhood cancer: a population-based cohort study

176. Life after childhood cancer: a population-based cohort study

177. Life after childhood cancer: a population-based cohort study

178. Life after childhood cancer: a population-based cohort study

179. Life after childhood cancer: a population-based cohort study

180. Life after childhood cancer: a population-based cohort study

181. Life after childhood cancer: a population-based cohort study

182. Life after childhood cancer: a population-based cohort study

183. Life after childhood cancer: a population-based cohort study

184. Life after childhood cancer: a population-based cohort study

185. Life after childhood cancer: a population-based cohort study

186. Life after childhood cancer: a population-based cohort study

187. Life after childhood cancer: a population-based cohort study

188. Life after childhood cancer: a population-based cohort study

189. Life after childhood cancer: a population-based cohort study

190. Life after childhood cancer: a population-based cohort study

191. Life after childhood cancer: a population-based cohort study

192. Life after childhood cancer: a population-based cohort study

193. Life after childhood cancer: a population-based cohort study

194. Life after childhood cancer: a population-based cohort study

195. Life after childhood cancer: a population-based cohort study

196. Life after childhood cancer: a population-based cohort study

197. Life after childhood cancer: a population-based cohort study

198. Life after childhood cancer: a population-based cohort study

199. Life after childhood cancer: a population-based cohort study

200. Life after childhood cancer: a population-based cohort study

201. Life after childhood cancer: a population-based cohort study

202. Life after childhood cancer: a population-based cohort study

203. Life after childhood cancer: a population-based cohort study

204. Life after childhood cancer: a population-based cohort study

205. Life after childhood cancer: a population-based cohort study

206. Life after childhood cancer: a population-based cohort study

207. Life after childhood cancer: a population-based cohort study

208. Life after childhood cancer: a population-based cohort study

209. Life after childhood cancer: a population-based cohort study

210. Life after childhood cancer: a population-based cohort study

211. Life after childhood cancer: a population-based cohort study

212. Life after childhood cancer: a population-based cohort study

213. Life after childhood cancer: a population-based cohort study

214. Life after childhood cancer: a population-based cohort study

215. Life after childhood cancer: a population-based cohort study

216. Life after childhood cancer: a population-based cohort study

217. Life after childhood cancer: a population-based cohort study

218. Life after childhood cancer: a population-based cohort study

219. Life after childhood cancer: a population-based cohort study

220. Life after childhood cancer: a population-based cohort study

221. Life after childhood cancer: a population-based cohort study

222. Life after childhood cancer: a population-based cohort study

223. Life after childhood cancer: a population-based cohort study

224. Life after childhood cancer: a population-based cohort study

225. Life after childhood cancer: a population-based cohort study

226. Life after childhood cancer: a population-based cohort study

227. Life after childhood cancer: a population-based cohort study

228. Life after childhood cancer: a population-based cohort study

229. Life after childhood cancer: a population-based cohort study

230. Life after childhood cancer: a population-based cohort study

231. Life after childhood cancer: a population-based cohort study

232. Life after childhood cancer: a population-based cohort study

233. Life after childhood cancer: a population-based cohort study

234. Life after childhood cancer: a population-based cohort study

235. Life after childhood cancer: a population-based cohort study

236. Life after childhood cancer: a population-based cohort study

237. Life after childhood cancer: a population-based cohort study

238. Life after childhood cancer: a population-based cohort study

239. Life after childhood cancer: a population-based cohort study

240. Life after childhood cancer: a population-based cohort study

241. Life after childhood cancer: a population-based cohort study

242. Life after childhood cancer: a population-based cohort study

243. Life after childhood cancer: a population-based cohort study

244. Life after childhood cancer: a population-based cohort study</



End of 1st symposium

Thank You

Linkedin connect!

